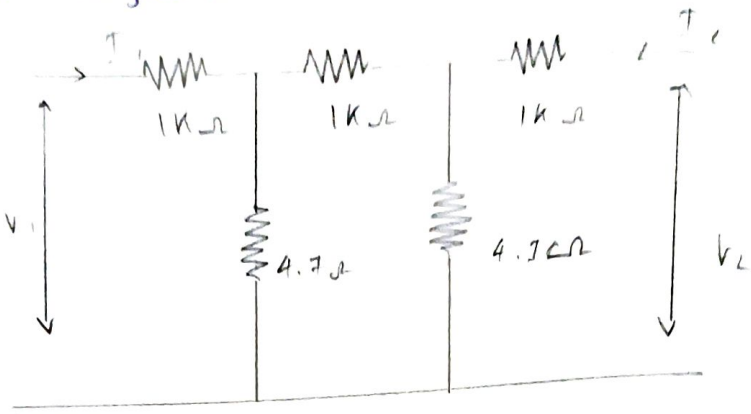
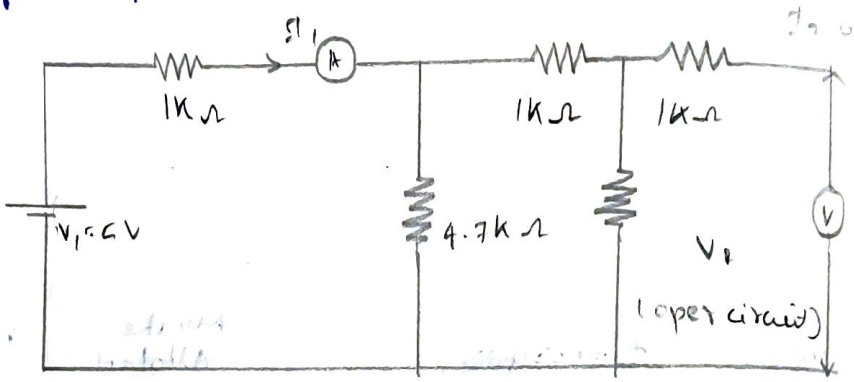
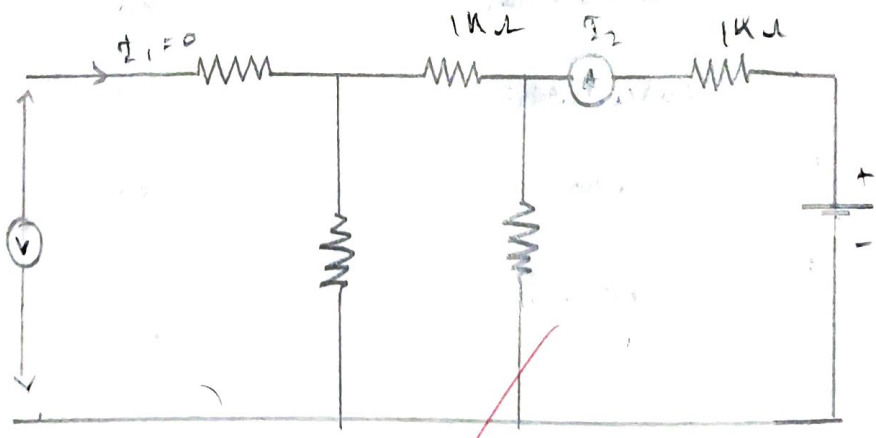
circuit Diagram.




output open circuited:




Input open circuited:



Exp No: 9

Date: 07/03/2026

# Determination of z parameters of Two Port Network.

Aim:

* To calculate and verify z parameters of a two port network using multisim simulator

Apparatus Required.

| S.NO | Components | Specification | Quantity |
|---|---|---|---|
| 1. | Power supply | – | 1 |
| 2. | Multimeter | – | 1 |
| 3. | Resistor | 10k, 10k, 4.7k | |
| 4. | Bread Board | – | 1 |
| 5. | Connecting wires | – | 1 |
| 6. | PC with resti multisim simulator | – | – |

Observation:

| S.NO | output open circuited $I_2 = 0$ | | | $Z_{21} = V_2/I_1$ | $Z_{11} = V_1/I_1$ |
|---|---|---|---|---|---|
| | $V_1$ | $V_2$ | $I_1$ | $\Omega$ | |
| 1. | 5 V | 2.97 V | 1.39 mA | 2.136 $\Omega$ | 3.597 $\Omega$ |

| S.NO | output open circuited $I_1 = 0$ | | | $Z_{21} = V_2/I_1$ | $Z_{11} = V_1/I_1$ |
|---|---|---|---|---|---|
| | $V_1$ | $V_2$ | $I_2$ | ($\Omega$) | ($\Omega$) |
| 1. | 2.97 V | 5 V | 1.39 mA | 2.136 $\Omega$ | 3.597 $\Omega$ |

Procedure:

* connect the circuit as shown in fig and switch 'on' the experiment

* First open the o/p terminal and supply 5V to I/p terminal measure o/p voltage and I/p current.

* Secondly, open I/p terminal and supply 5V to o/p terminal Measure I/p voltage and o/p current using multimeter

* Calculate the values of 2 parameters using equation 1 and 2

* switch "off" the supply after taking the readings.

calculation

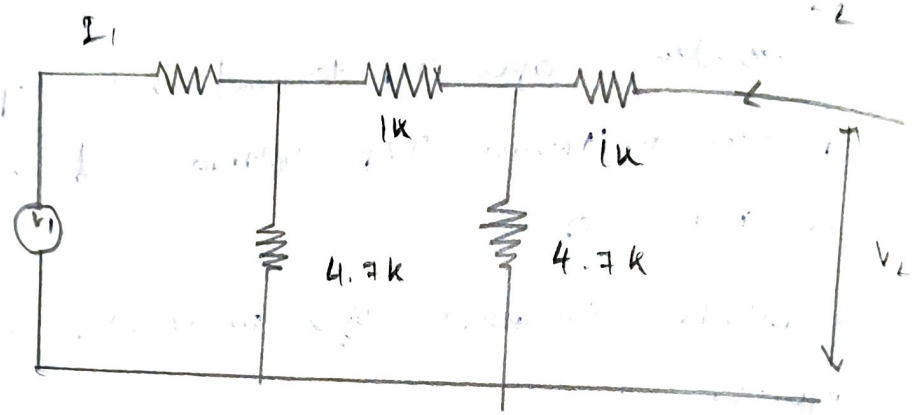
theortical calculation

to find $z_1$ & $z_{21}$

$$V_1 = z_1 I_1 + z_{12} I_2$$

$$V_2 = z_{21} I_1 + z_{22} I_2$$

when $I_2 = 0$ => port 2 open circuit




$$R_{eq} = 1 + 4.7 = 5.7$$

$$R_{eq_n} = \frac{5.7 \times 4.7}{5.7 + 4.7}$$

$$= 2.575$$

$$R = 2.575 + 1 = 3.575 k\Omega$$

By Ohm's Law

$$V_1 = I R$$

$$\frac{V_1}{I_1} = R$$

$$z_{11} = \frac{V_1}{I} = 3.575 k\Omega$$

By current Division Rule

$$I_2 = I_1 \times \frac{4.7}{4.7+1+4.7}$$

$$I_2 = \frac{4.7 I_1}{10.4}$$

$$I_2 = 0.451 I$$

ohm's law,

$$V_2 = I_2 \times 4.7$$

$$V_2 = 4.7 \times I_1 \times 0.451$$

$$\frac{V_2}{I_1} = 2.1197$$

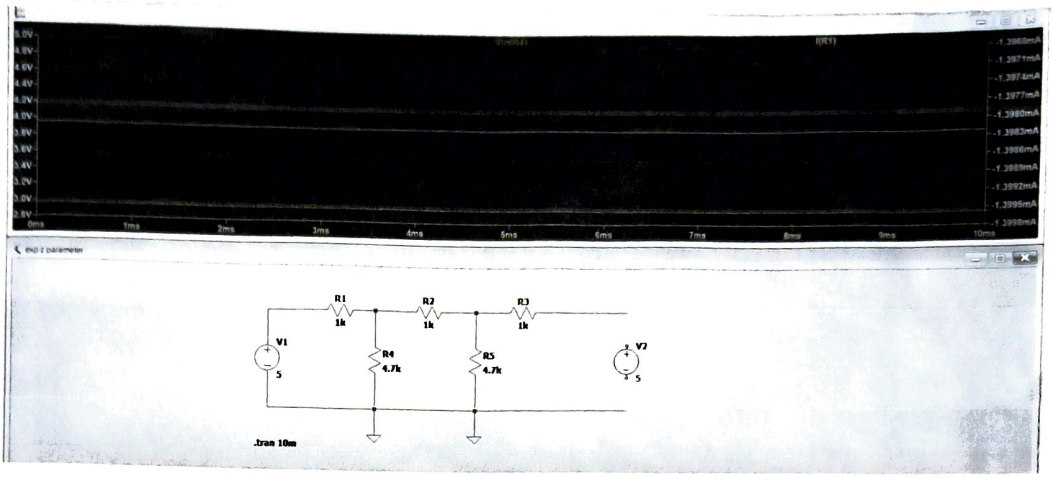
$$Z_{21} = \frac{V_2}{I_1} = 2.1197 k\Omega$$

$\therefore$ The network is symmetrical

$$Z_{11} = Z_{22}$$

$$Z_{21} = Z_{12}$$

$$Z = \begin{bmatrix} 3.575 & 2.1197 \\ 2.1197 & 2.575 \end{bmatrix}$$




| S.NO | Description | Marks Allotted | Marks Obtained |
|---|---|---|---|
| 1. | Experimental setup | 20 | 18 |
| 2. | Observation | 20 | 18 |
| 3. | calculation | 20 | 18 |
| 4. | viva | 20 | 18 |
| 5. | Record | 20 | 20 |
| | | | 92 |

Result:

* the z-parameters of the two port network has been calculated and verified by using multisim simulator.